



Saint Ignatius' College, Riverview

Mathematics Assessment Task

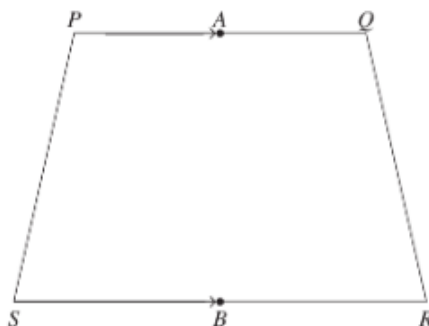
2021

Year 12
Mathematics (Extension One)
Task 4
Date: 20 th August 2021

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Question 1 (START A NEW WRITING PAGE)

- (a) Consider the polynomial $P(x) = x^3 - x^2 - 8x + 12$.
- (i) Use the factor theorem to show that $(x - 2)$ is a factor of $P(x)$. (1)
- (ii) Factorise $P(x)$ completely. (2)
- (b) Solve the equation $\sin 2x = \tan x$ for the domain $[0, \pi)$ (3)
- (c) Prove $n(n + 2)$ is divisible by 4 by mathematical induction, if n is any positive **even** integer. (3)
- (d) Find the value of k if $y = \tan x$ satisfies the differential equation $y' = k + y^2$. (2)
- (e) PQRS is a trapezium with A and B being the midpoints of PQ and RS respectively.



Let $\overline{PA} = a$ and $\overline{SB} = b$

- (i) Show that $\overline{QR} = b - a + \overline{AB}$ (1)
- (ii) Express $\overline{PS}$ in terms of a , b and $\overline{AB}$. (1)
- (iii) Write an expression for $\overline{PS} + \overline{QR}$, in terms of $\overline{AB}$ (2)

END OF QUESTION 1

QUESTION 2 (START A NEW WRITING PAGE)

- (a) (i) Write the expression $\sin x + \sqrt{3} \cos x$ in the form $R \sin(x + \alpha)$, where $R > 0$. (2)
and $0 \leq x \leq \frac{\pi}{2}$

- (ii) Hence, solve the equation $\sin x + \sqrt{3} \cos x = 1$ in the domain $[0, 2\pi]$ (2)

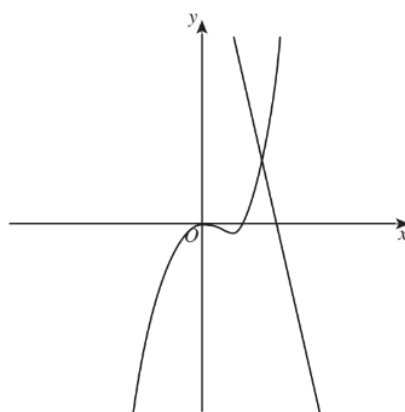
- (b) After time t years from the start of the year 2021, the number of people in a population is given by:

$$N = 70 + Ae^{0.1t} \quad \text{where } A \text{ is a constant greater than } 0.$$

- (i) Show that $\frac{dN}{dt} = 0.1(N - 70)$. (1)

- (ii) There were 100 people in the population at the start of the year 2021. (3)
Find the year when the population size will exceed 190.

- (c) The graphs of the functions $g(x) = mx + b$ and $f(x) = 3x^3 - 2x^2$ are shown.



- (i) Sketch the graph of $y = 3|x|^3 - 2|x|^2$ (1)

- (ii) Use the sketch of $y = 3|x|^3 - 2|x|^2$, to find the values of m and b if the function of $g(x) = mx + b$, has the solution to the equation $mx + b < 3|x|^3 - 2|x|^2$ is $x < -3$ or $x > 1$. (2)

QUESTION 2 continues the next page...

QUESTION 2 (continued)

- (d) A NSW census shows that 40% of NSW adults completed at least 30 minutes of exercise each day.
- (i) A random sample of 21 NSW adults is to be conducted to find the proportion who completed at least 30 minutes of exercise each day. Given that the mean is 0.4, show that standard deviation for the distribution of sample proportions 0.1069. (2)
- (ii) Part of a table giving values of $P(Z \leq z)$, where z is a standard normal variable is shown below (2)

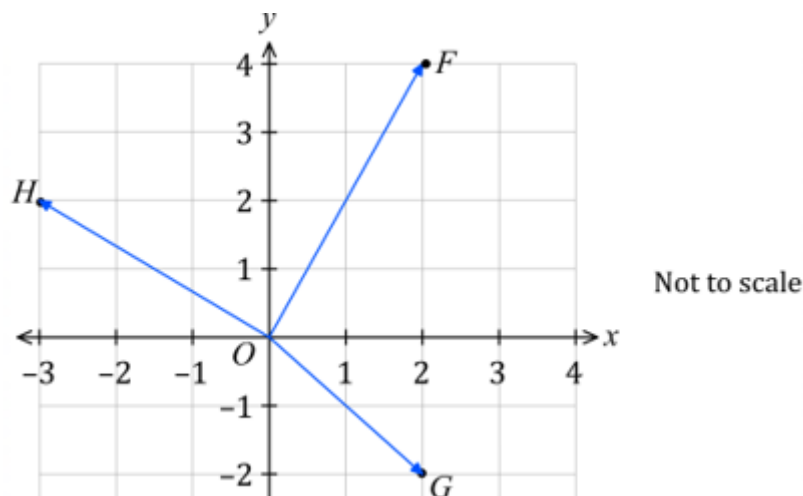
z	+0.00	+0.01	+0.02	+0.03	+0.04	+0.05	+0.06	+0.07	+0.08	+0.09
2.4	0.9918	0.9920	0.9922	0.9925	0.9927	0.9929	0.9931	0.9931	0.9934	0.9936
2.5	0.9938	0.9940	0.9943	0.9945	0.9945	0.9946	0.9948	0.9949	0.9951	0.9952
2.6	0.9953	0.9955	0.9956	0.9957	0.9959	0.9960	0.9961	0.9962	0.9963	0.9964

Estimate the probability that a random sample of 21 NSW adults find at most 3 who have completed at least 30 minutes of exercise each day.

END OF QUESTION 2

QUESTION 3 (START A NEW WRITING PAGE)

- (a) The vectors $\overrightarrow{OF}$, $\overrightarrow{OG}$ and $\overrightarrow{OH}$ are shown below.



What is the size of $\angle GOH$ to the nearest degree? (2)

- (b) What is the derivative of $\sin x \cdot \cos^{-1} x$? (2)

- (c) Use the substitution $u = \cos 2\theta$ to evaluate $\int_{\frac{\pi}{2}}^{\frac{3\pi}{4}} \cos^2 2\theta \sin 2\theta \, d\theta$. (2)

- (d) A spherical balloon is to be filled with water so that its surface area increases at a constant rate of $1 \text{ cm}^2/\text{s}$.

[Note: Sphere formulae $SA = 4\pi r^2$ and $V = \frac{4}{3}\pi r^3$]

- (i) Find when the radius is 3 cm:

(α) the required rate of increase of the radius; (2)
(leave your answer in terms of π)

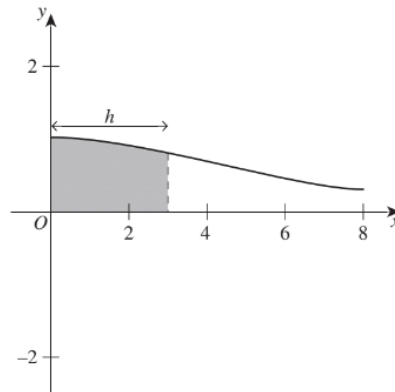
(β) the rate at which the water is flowing in at that time. (1)

- (ii) Find the volume when the volume is increasing at $10 \text{ cm}^3/\text{s}$ (2)
(leave your answer in terms of π)

QUESTION 3 continues the next page...

QUESTION 3 continued...

- (e) The graph of the function $y = \frac{3}{\sqrt{9+x^2}}$ for $x > 0$ is shown.



- The area bounded by the curve $y = \frac{3}{\sqrt{9+x^2}}$, the axes and the lines $x=0$ and $x=h$ is rotated about the x -axis to create a solid of revolution. (2)

Find in terms of h , the volume of this solid.

- (f) The spread of flu through a student population is modelled by the equation

$$S = \frac{2000}{1 + 199e^{-0.4t}}$$

where S is the total number of students infected after t days.

- Show that the given equation for S satisfies the differential equation (2)

$$\frac{dS}{dt} = \frac{S}{5} \left(2 - \frac{S}{1000} \right).$$

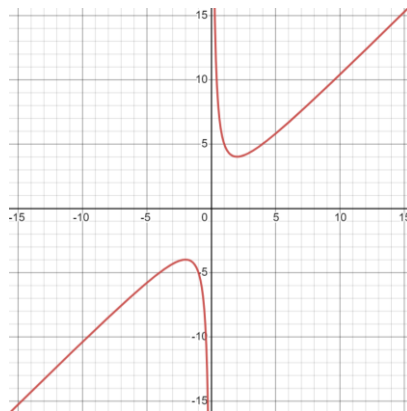
END OF QUESTION 3

QUESTION 4 (START A NEW WRITING PAGE)

(a) Show that $\int_0^{\frac{\pi}{4}} (1 + \tan x)^2 dx = \ln 2 + 1$ (3)

(b) Prove that $\frac{\cos 3\theta}{\cos \theta} - \frac{\sin 3\theta}{\sin \theta}$ is independent of θ . (2)

(c) The graph of the function $f(x) = \frac{x^2 + 4}{x}$ is shown.



(i) Show that the stationary points of $f(x) = \frac{x^2 + 4}{x}$ are $(-2, -4)$ and $(2, 4)$. (2)

(ii) Sketch the graph of $y = \frac{1}{\sqrt{f(x)}}$, (2)

showing all important features including turning point(s), intercept(s) and asymptotes.

QUESTION 4 continues the next page...

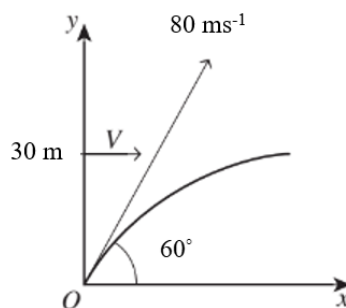
QUESTION 4 continued ...

- (d) Initially, a golf ball was hit from the ground at a velocity of 80 ms^{-1} at an angle of 60° to the horizontal and $g = 10 \text{ m s}^{-2}$.

The position vector of the golf ball after t seconds is given by

$$\underline{s}(t) = (40t)\underline{i} + (40\sqrt{3}t - 5t^2)\underline{j}.$$

- (i) Eight seconds **after** the golf ball was hit, a small stone was fired at a velocity V **horizontally** from a point 30 m above the ground.



- Find the position vector of the stone. (2)
- (ii) Find the time to the nearest second after the golf ball was hit when it collided with the stone. (2)
- (iii) What is the stone's speed at collision? (2)
- (Give your answer to three significant figures)

END OF QUESTION 4

END OF EXAMINATION



SUGGESTED SOLUTIONS

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Q1 (a) $P(x) = x^3 - x^2 - 8x + 12$

(i) $P(2) = 2^3 - 2^2 - 8(2) + 12$ } either seen 1 Mark
 $= 8 - 4 - 16 + 12$
 $= 0$

$\therefore (x-2)$ is a factor.

(ii) $P(-3) = (-3)^3 - (-3)^2 - 8(-3) + 12$
 $= -27 - 9 + 24 + 12$
 $= 0$

Various methods available

$\therefore (x+3)$ is a factor.

Now, $x \div 8 = -12$
 $2(-3) \div 8 = -12$
 $8 = 2$

Mark 2. CAO
 Mark 1. Second factor identified.

$\therefore P(x) = (x-2)^2(x+3)$

(b) $\sin 2x = \tan x$ $[0, \pi)$

$2 \sin x \cos x = \frac{\sin x}{\cos x}$

$\therefore 2 \sin x \cos^2 x - \sin x = 0$

$\sin x (2 \cos^2 x - 1) = 0$

$\therefore \sin x = 0$ OR $2 \cos^2 x - 1 = 0$

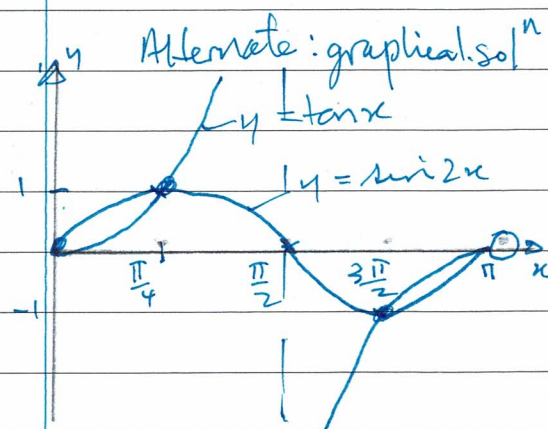
$x = 0$

$\cos^2 x = \frac{1}{2}$

$\cos x = \pm \frac{1}{\sqrt{2}}$

$x = \frac{\pi}{4}, \frac{3\pi}{4}$

$x = 0, \frac{\pi}{4}, \frac{3\pi}{4}$



Mark 3: Three correct solutions

Mark 2: Considerable progress

Mark 1: Some progress made.

(c) $n(n+2) = 4N$ (N an integer)

1. Prove true for $n=2$

$2(2+2) = 4 \times 2 \therefore$ true for $n=2$

2. Assume true for $n=k$

i.e. $k(k+2) = 4N$ (N is an integer)

3. Prove true for $n=k+2$

i.e. $(k+2)(k+2+2) = 4P$ (P an integer)

LHS = $(k+2)(k+4)$

$= k(k+2) + 4(k+2)$

(continued over)

Mark 1: Steps 1+2

Mark 2: Set up in $k+2$

Mark 3: Substitution and resolution

$$LHS = K(K+2) + 4(K+2)$$

$$= 4N + 4(K+2)$$

$$= 4(N+K+2) \text{ as } N, K \text{ and } 2$$

$$= 4P$$

$$= RHS$$

are all integers

$(N+K+2)$ is an

$\therefore$ true for $n=K+2$ if true integer.

for $n=K$. As true for

$n=2$, true for $n=4$ and so on.

True for all positive even numbers.

Problem 1: let $n=2K$

many did not define K . If K is even

then even integers are missed

- leaving the substitution with K in the denominator

is the term an integer e.g. $\frac{8}{6} = \frac{4}{3}$

a non-integer.

Mark 1: $y' = \sec^2 x$

$$(d) y = \tan x$$

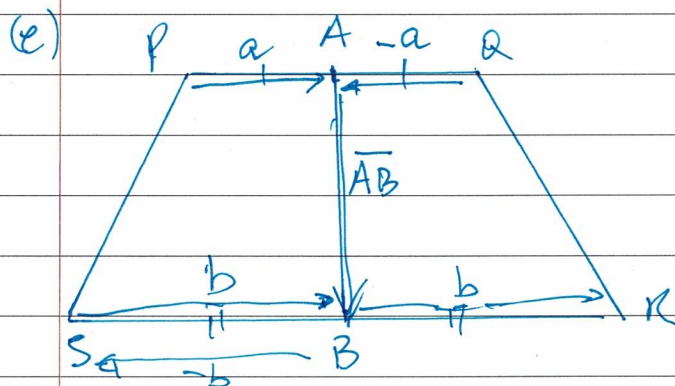
$$y' = \sec^2 x$$

$$\therefore \sec^2 x = K + \tan^2 x$$

$$\text{Pythag: } \sec^2 x = 1 + \tan^2 x$$

$$\therefore K = 1$$

$$\text{Mark 2: } K = 1$$



YOU SHOULD HAVE A DIAGRAM AS PART OF YOUR ANSWER!!

$$(i) \overline{QR} = -\overline{a} + \overline{AB} + \overline{b}$$

$$= \underbrace{\overline{b}} - \underbrace{\overline{a}} + \overline{AB}$$

$$(ii) \overline{PS} = \underbrace{\overline{a}} + \overline{AB} - \underbrace{\overline{b}}$$

$$= -\underbrace{\overline{b}} + \underbrace{\overline{a}} + \overline{AB}$$

$$(iii) \overline{PS} + \overline{QR} = -\underbrace{\overline{b}} + \underbrace{\overline{a}} + \overline{AB} + \underbrace{\overline{b}} - \underbrace{\overline{a}} + \overline{AB}$$

$$= 2\overline{AB}$$

$$(i) \text{ Diagram or notation}$$

$$\overline{QR} = \overline{QA} + \overline{AB} + \overline{BR}$$

$$= -\overline{a} + \overline{AB} + \overline{b}$$

1 Mark

Mark 2: $2\overline{AB}$
Mark 1: attempted addition

QUESTION TWO

$$a) i) \sin x + \sqrt{3} \cos x = R \sin(x + \alpha)$$

$$\text{Now: } R \sin(x + \alpha) = R \sin x \cos \alpha + R \cos x \sin \alpha$$

$$\therefore \sin x + \sqrt{3} \cos x = R \sin x \cos \alpha + R \cos x \sin \alpha$$

$$\text{If } R \cos \alpha = 1 \quad \therefore \tan \alpha = \sqrt{3}$$

$$R \sin \alpha = \sqrt{3}$$

$$\alpha = \frac{\pi}{3} \leftarrow \text{link}$$

$$R^2 \cos^2 \alpha + R^2 \sin^2 \alpha = 1 + 3$$

$$R^2 (\cos^2 \alpha + \sin^2 \alpha) = 4 \quad \therefore R = 2 \leftarrow \text{link}$$

$$\text{So } \sin x + \sqrt{3} \cos x = 2 \sin \left(x + \frac{\pi}{3}\right)$$

$$ii) 2 \sin \left(x + \frac{\pi}{3}\right) = 1 \quad [0, 2\pi]$$

$$\sin \left(x + \frac{\pi}{3}\right) = \frac{1}{2}$$

$$\therefore x + \frac{\pi}{3} = \frac{\pi}{6}, \frac{5\pi}{6}, \frac{13\pi}{6}, \frac{17\pi}{6} \dots$$

$$x = -\frac{\pi}{6}, \frac{\pi}{2}, \frac{11\pi}{6}, \dots$$

$$\therefore \text{For } [0, 2\pi]$$

$$x = \frac{\pi}{2}, \frac{11\pi}{6} \leftarrow \text{link for each}$$

Well done

Working was needed.

Well done.

$$b) i) N = 70 + Ae^{0.1t}$$

$$\frac{dN}{dt} = 0.1 Ae^{0.1t}$$

$$\text{Note: } Ae^{0.1t} = N - 70$$

} 1mk for EITHER.

$$\therefore \frac{dN}{dt} = 0.1(N - 70)$$

$$ii) N = 70 + A e^{0.1t}$$

$$t=0 \quad N = 100$$

$$\therefore 100 = 70 + A e^0$$

$$\therefore A = 30 \leftarrow 1mk$$

$$N = 70 + 30 e^{0.1t}$$

$$\text{Let } N = 190$$

$$190 = 70 + 30 e^{0.1t} \leftarrow 1mk$$

$$4 = e^{0.1t}$$

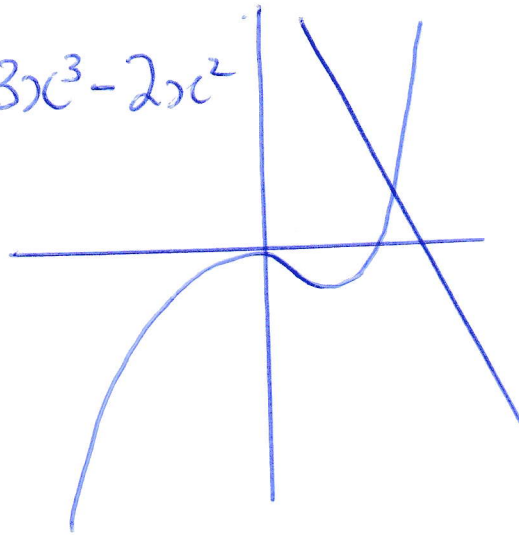
$$\therefore t = \frac{\ln 4}{0.1}$$

$$= 13.8 \text{ yrs}$$

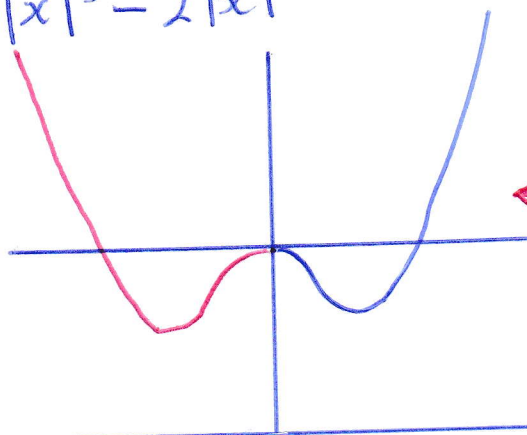
$$\text{ie Year 2034} \leftarrow 1mk$$

c)

$$y = 3x^3 - 2x^2$$



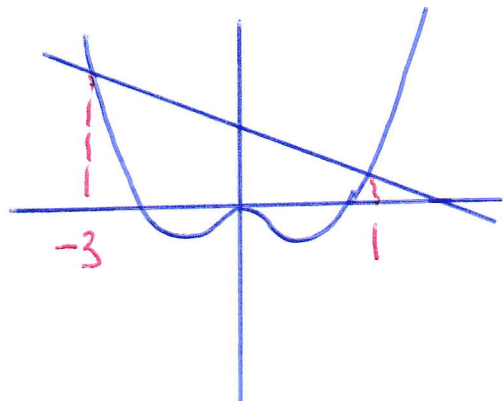
i) $y = 3|x|^3 - 2|x|^2$



Many struggled.

← link

ii)



$$x = -3 \quad (-3, 63)$$

$$y = 63$$

$$x = 1 \quad (1, 1)$$

$$y = 1$$

$$m = \frac{63 - 1}{-3 - 1}$$

$$= -\frac{31}{2}$$

$$= -15.5$$

$$y = -\frac{31}{2}x + b$$

$$(1, 1)$$

$$1 = -\frac{31}{2} + b$$

$$b = \frac{33}{2}$$

$$= 16.5$$

Poorly done

link for correct m =
and b =

"Part marks awarded for some progress"

d) i) $p = 0.4$ $n = 21$

$$\sigma = \sqrt{\text{VAR}}$$

$$= \sqrt{\frac{0.4(1-0.4)}{21}}$$

$$\approx 0.1069$$

A number of methods used.

2mks

ii) $Z = \frac{x - \mu}{\sigma}$ $x = \frac{3}{21}$
 $= 0.1428$

$$Z = \frac{0.1428 - 0.4}{0.1069}$$

$$= -2.41$$

1mk

A z score at 2.41
 gives a probability at
 0.9920

Too many misread the table

∴ for $Z = -2.41$

$$\text{Prob} = 0.008$$

1mk

Yr 12ExtensionOneMathsExam(Trial)

Q3.)

$$(a) \quad \vec{OH} = -3\hat{i} + 2\hat{j}$$

$$|\vec{OH}| = \sqrt{13}$$

$$\vec{OG} = 2\hat{i} - 2\hat{j}$$

$$|\vec{OG}| = \sqrt{8}$$

$$\frac{\vec{OH} \cdot \vec{OG}}{|\vec{OH}| |\vec{OG}|} = \cos \angle GOH$$



(Well Answered)

$$\frac{-6 - 4}{\sqrt{13} \sqrt{8}} = \cos \angle GOH$$

$$\angle GOH = \cos^{-1} \frac{-10}{\sqrt{104}} = 168.69^\circ = 169^\circ$$



b)

$$y = \sin x \cos^{-1} x$$

$$y = u \times v$$

$$y' = u v' + v u'$$

$$y' = (\sin x) \left(\frac{-x}{\sqrt{1-x^2}} \right) + \cos x \cos^{-1} x$$

$$y' = \frac{-\sin x}{\sqrt{1-x^2}} + \cos x \cos^{-1} x$$

(Well Answered)

Some students changed $\cos x \cos^{-1} x$ to x or 1 which is incorrect

$$c) \quad u = \cos 2\theta$$

$$\theta = \frac{3\pi}{4}$$

$$u = \cos \frac{3\pi}{2} = 0$$

$$\frac{du}{d\theta} = -2 \sin 2\theta$$

$$\theta = \frac{\pi}{2}$$

$$u = \cos \pi = -1$$

$$\left(\begin{array}{l} du = -2 \sin 2\theta d\theta \\ \text{* Some students forget to include 2 in} \\ \text{derivative here} \end{array} \right) \quad -\frac{1}{2} du = \sin 2\theta d\theta$$

$$I = -\frac{1}{2} \int_{-1}^0 u^2 du$$

(Well Answered)

$$I = \frac{1}{2} \int_0^{-1} u^2 du$$

$$I = \frac{1}{2} \left[\frac{u^3}{3} \right]_0^{-1}$$

$$I = \cancel{0} = -\frac{1}{6}$$

(Some students made mistakes calculating in the last step)

d)

(i)

$$a) \quad \frac{dA}{dt} = 1 \text{ cm}^2/\text{s}$$

$$\frac{dr}{dt} = ? \quad \text{when } r = 3 \text{ cm}$$

$$\frac{dr}{dt} = \frac{dA}{dt} \times \frac{dr}{dA}$$



$$A = 4\pi r^2$$

$$\frac{dA}{dr} = 8\pi r$$

$$\frac{dr}{dA} = \frac{1}{8\pi r}$$

(Well answered)

$$\frac{dr}{dt} = 1 \times \frac{1}{8\pi r} = \frac{1}{8\pi r} \text{ cm/s} = \frac{1}{24\pi}$$

6) $\frac{dV}{dt} = ?$

$$r = 3 \text{ cm}$$

$$\frac{dV}{dt} = \frac{dV}{dr} \times \frac{dr}{dt}$$

(Well answered)

$$V = \frac{4}{3} \pi r^3$$

$$\frac{dV}{dr} = 3 \left(\frac{4}{3} \right) \pi r^2 = 4\pi r^2 \quad \text{when } r = 3$$

$$\frac{dV}{dr} = 36\pi$$

$$\frac{dV}{dt} = 36\pi \times \frac{1}{8\pi(3)} = \frac{36}{24} = \frac{3}{2} = 1.5 \text{ cm}^3/\text{s}$$

(ii) $\frac{dV}{dt} = \frac{dV}{dr} \times \frac{dr}{dt}$

(Many students assumed $r=3$ here)

$$10 = 4\pi r^2 \times \frac{1}{8\pi r}$$

$$10 = \frac{4r^2}{8r} \quad \checkmark$$

$$V = \frac{4}{3} \pi (20)^3 = \frac{32000}{3} \pi \text{ cm}^3 \quad \checkmark$$

$$80 = 4r$$

$$r = 20$$

$$e) \quad y = \frac{3}{\sqrt{9+x^2}}$$

$$V = \pi \int_0^h y^2 dx$$

$$V = \pi \int_0^h \frac{9}{9+x^2}$$

$$V = 9\pi \int_0^h \frac{1}{3^2+x^2} \quad \checkmark$$

$$V = 9\pi \left[\frac{1}{3} \tan^{-1} \frac{x}{3} \right]_0^h$$

$$V = 9\pi \left(\frac{1}{3} \tan^{-1} \frac{h}{3} \right) - 9\pi \left(\frac{1}{3} \tan^{-1} 0 \right)$$

$$V = 3\pi \tan^{-1} \frac{h}{3} \quad \checkmark$$

(Well Answered)



$$f) \quad S = \frac{2000}{1 + 199e^{-0.4t}}$$

L.H.S

R.H.S.

$$\frac{dS}{dt}$$

$$\frac{S}{5} \left(2 - \frac{S}{1000} \right) =$$

$$S = \frac{2000}{1 + 199e^{-0.4t}}$$

$$\frac{S}{5} \left(\frac{2000 - S}{1000} \right) =$$

$$S = 2000(1 + 199e^{-0.4t})^{-1}$$

$$\frac{S(2000 - S)}{5000}$$

$$\frac{dS}{dt} = -2000(1 + 199e^{-0.4t})^{-2} (-0.4 \times 199 \times e^{-0.4t})$$

$$\frac{dS}{dt} = \frac{(-2000)(-79.6e^{-0.4t})}{(1 + 199e^{-0.4t})^2}$$

$$\frac{2000}{1 + 199e^{-0.4t}} \left(\frac{2000 - 2000}{1 + 199e^{-0.4t}} \right)$$

$$\frac{dS}{dt} = \frac{159200e^{-0.4t}}{(1 + 199e^{-0.4t})^2}$$

$$\frac{2000}{1 + 199e^{-0.4t}} \left(\frac{2000(1 + 199e^{-0.4t}) - 2000}{1 + 199e^{-0.4t}} \right)$$

(Poorly Answered)

$$\frac{2000^2 + 2000(199e^{-0.4t}) - 2000}{(1 + 199e^{-0.4t})^2}$$

(You need to work down L.H.S & R.H.S separately and show they are equal)

$$\frac{2000(2000(1 + 199e^{-0.4t}) - 2000)}{5000(1 + 199e^{-0.4t})^2}$$

$$= \frac{(2000)^2 199e^{-0.4t}}{5000(1 + 199e^{-0.4t})^2}$$

$$= \frac{159200e^{-0.4t}}{(1 + 199e^{-0.4t})^2}$$

Q4

$$\begin{aligned}
 (a) \quad & \int_0^{\frac{\pi}{4}} (1 + \tan x)^2 dx \\
 &= \int_0^{\frac{\pi}{4}} 1 + 2 \tan x + \tan^2 x dx \\
 &= \int_0^{\frac{\pi}{4}} 1 + 2 \frac{\sin x}{\cos x} + \sec^2 x - 1 dx \\
 &= 2 \int_0^{\frac{\pi}{4}} \frac{\sin x}{\cos x} + \sec^2 x dx \quad \checkmark \text{ --- 1 Mark} \\
 &= \left[-2 \ln(\cos x) + \tan x \right]_0^{\frac{\pi}{4}} \quad \checkmark \text{ --- 1 Mark} \\
 &= -2 \ln \frac{1}{\sqrt{2}} + 1 - (-2 \ln 1 + 0) \\
 &= 1 - 2 \ln 2^{\frac{1}{2}} \quad \checkmark \text{ --- 1 Mark} \\
 &= 1 + \ln 2
 \end{aligned}$$

(b)

$$\begin{aligned}
 & \frac{\cos 3\theta}{\cos \theta} - \frac{\sin 3\theta}{\sin \theta} \\
 &= \frac{\sin \theta \cos 3\theta - \cos \theta \sin 3\theta}{\cos \theta \sin \theta} \\
 &= \frac{\sin(-2\theta)}{\frac{1}{2} 2 \sin \theta \cos \theta} \quad \checkmark \text{ --- 1 Mark} \\
 &= \frac{-2 \sin 2\theta}{\sin 2\theta} \\
 &= -2 \quad \checkmark \text{ --- 1 Mark}
 \end{aligned}$$

(c) $f(x) = \frac{x^2 + 4}{x}$

(i)

$$f'(x) = \frac{x(2x) - (x^2 + 4)}{x^2}$$

$$= \frac{x^2 - 4}{x^2}$$

$$= 0 \text{ at SPs}$$

✓ ——— / mark

$$\therefore (x-2)(x+2) = 0$$

$$x = \pm 2$$

$$y = \pm 4$$

✓ ——— / mark

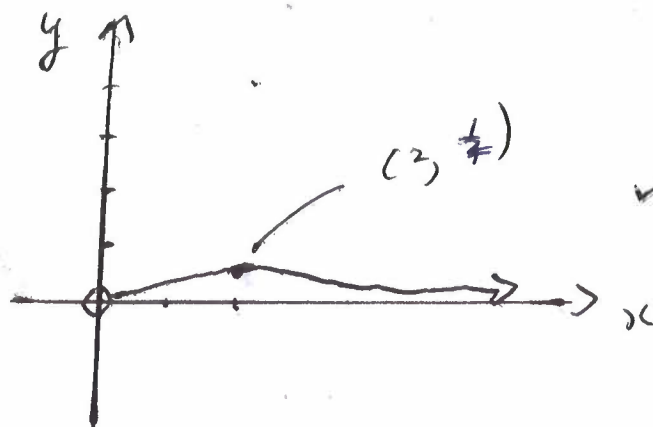
SPs are $(2, 4)$ and $(-2, -4)$

(ii)

$$y = \sqrt{f(x)}$$

$$\therefore f(x) > 0$$

✓ ——— / mark



✓ ——— correct
start point
identity

NOTE: should be
open circle at origin
but I accepted
open / closed

$$(d) \quad \underline{s}(t) = (40t) \underline{i} + (40\sqrt{3}t - 5t^2) \underline{j}$$

$$(i) \quad \underline{a}(t) = -10 \underline{j}$$

$$\underline{v}(t) = -10t \underline{j} + C$$

$$t=0 \quad \underline{v}(t) = V \underline{i}$$

$$\therefore C = V \underline{i}$$

$$\underline{v}(t) = V \underline{i} + (-10t) \underline{j} \quad \checkmark$$

$$\underline{s}(t) = Vt \underline{i} + (-5t^2) \underline{j} + C$$

$$t=0 \quad \underline{s}(t) = 30 \underline{j}$$

$$\therefore C = 30 \underline{j}$$

$$\underline{s}(t) = Vt \underline{i} + (-5t^2 + 30) \underline{j} \quad \checkmark$$

* NOTE!

Can also

have answer

as

$$V(t-8) \underline{i} + (-5(t-8)^2 + 30) \underline{j}$$

* Could also have

$$\underline{s}(t) = V(t-8) \underline{i} + (-5(t-8)^2 + 30) \underline{j}$$

(ii) when they collide

$$40t = V(t-8) \dots (1) \quad \text{and} \quad 40\sqrt{3}t - 5t^2 = 30 - 5(t-8)^2 \dots (2)$$

Using (2)

$$40\sqrt{3}t - 5t^2 = 30 - 5t^2 + 80t - 320 \quad \checkmark$$

$$290 = 80t - 40\sqrt{3}t$$

$$t = \frac{290}{80 - 40\sqrt{3}} \approx 27.057$$

$$\approx 27 \quad \checkmark$$

* NOTE! Poorly done as many students did not account for the time difference - no marks awarded

(iii) Need to find the initial velocity of stone, V .

from part (ii)

$$40t = V(t-8) \text{ when } t = 27$$

$$\therefore 40 \times 27 = V \times (27-8)$$

$$V = \frac{40 \times 27}{19} \doteq 56.84 \quad \checkmark \text{ — Mark}$$

$$\underline{V}(t) = V \underline{i} + (-10t) \underline{j}.$$

$$\underline{V}(t-8) = V \underline{i} + (-10(t-8)) \underline{j}.$$

Correct value
of V for
stone

$$\text{Speed} = \sqrt{V^2 + (-10(27-8))^2}$$

$$= \sqrt{\left(\frac{40 \times 27}{19}\right)^2 + (-10 \times 19)^2}$$

$$= 198.32$$

$$\doteq 198 \text{ m/s}$$

$\checkmark$ — correct
answer.